UNIVERSITY OF AMSTERDAM
Amsterdam Business School

Data Science in Online Marketing

Recommenders 1 – Content Based

Maarten Soomer

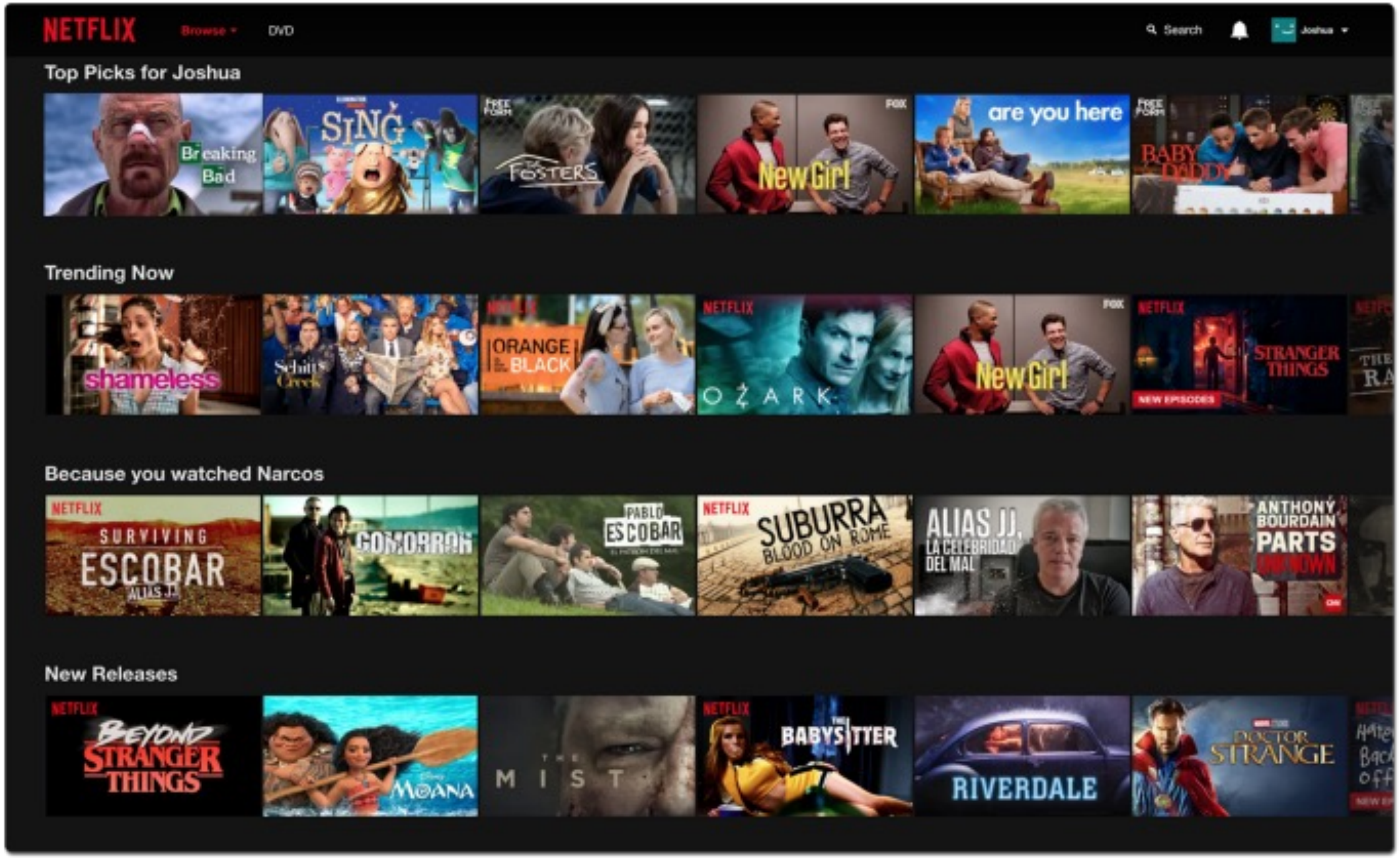
YouTube

*60% of the clicks on the home screen are on the recommendations**

* Measuring the Business Value of Recommender Systems, D. Jannach & M. Jugovac ACM Transactions on Management Information Systems, Volume 10, Issue 4, 2019

Image taken from: YouTube.com

Netflix

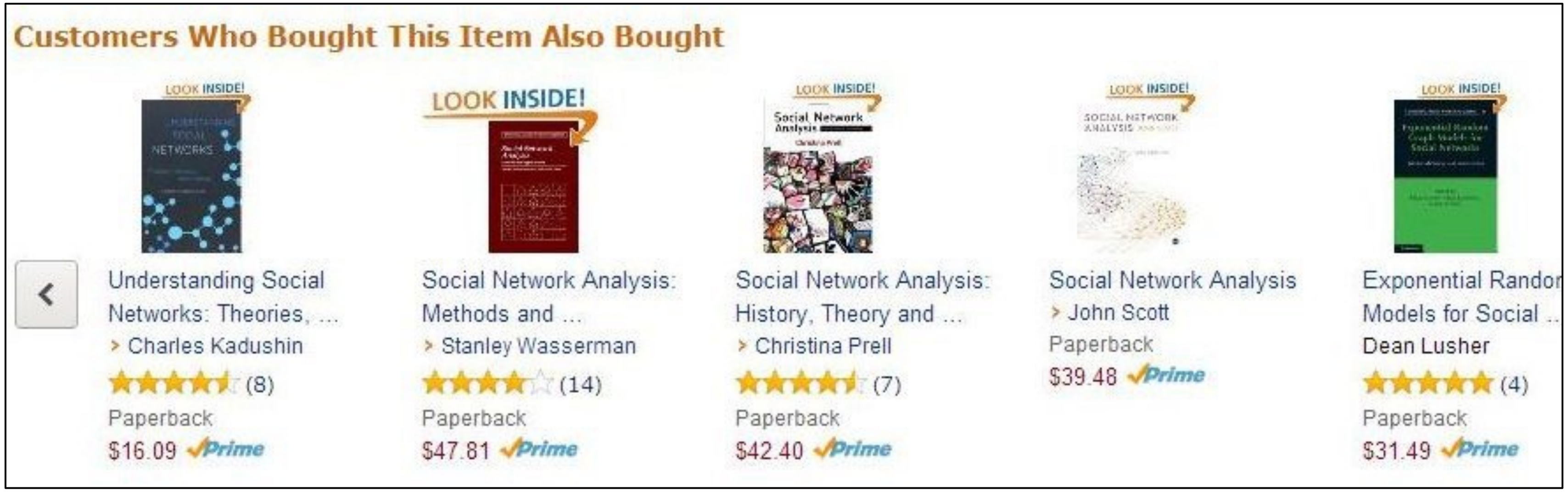


75 % of what people watch is from some sort of recommendation *

Image taken from: Netflix

* [Measuring the Business Value of Recommender Systems](), D. Jannach & M. Jugovac ACM Transactions on Management Information Systems, Volume 10, Issue 4, 2019

Amazon

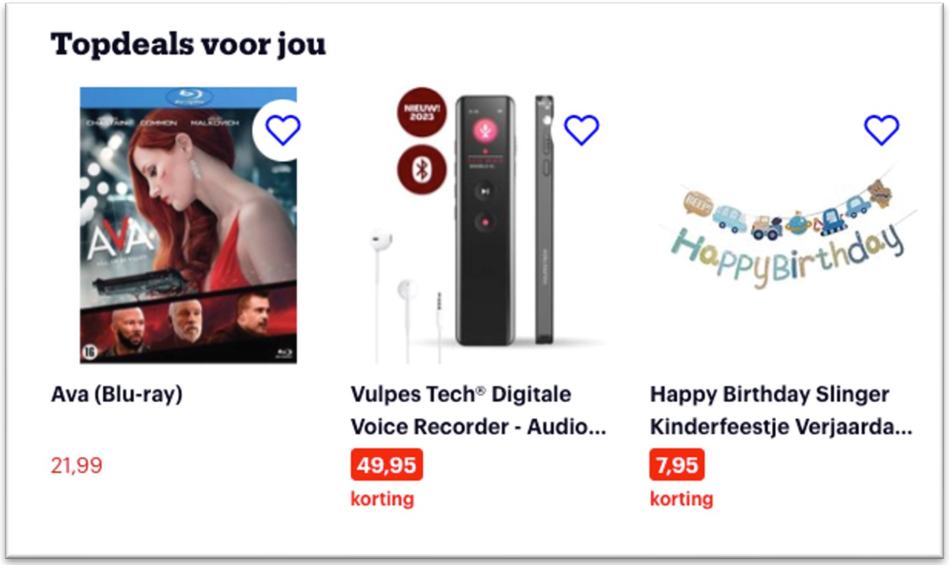


*Amazon reports that 35% of its sales comes from recommendations**

* Measuring the Business Value of Recommender Systems, D. Jannach & M. Jugovac ACM Transactions on Management Information Systems, Volume 10, Issue 4, 2019

Image taken from: Amazon.com

bol.com



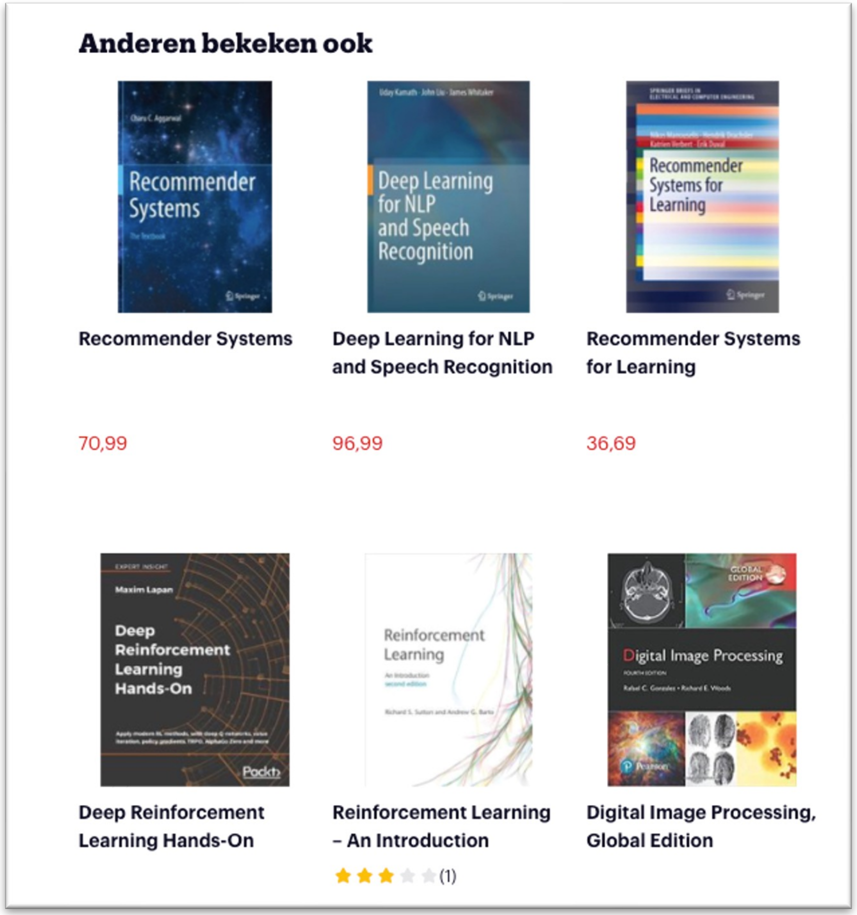


Image taken from: bol.com

Why Recommendations?

Value for customer

- Find what you’re looking for
- Narrow down the set of choices
- Discover new things

Value for provider

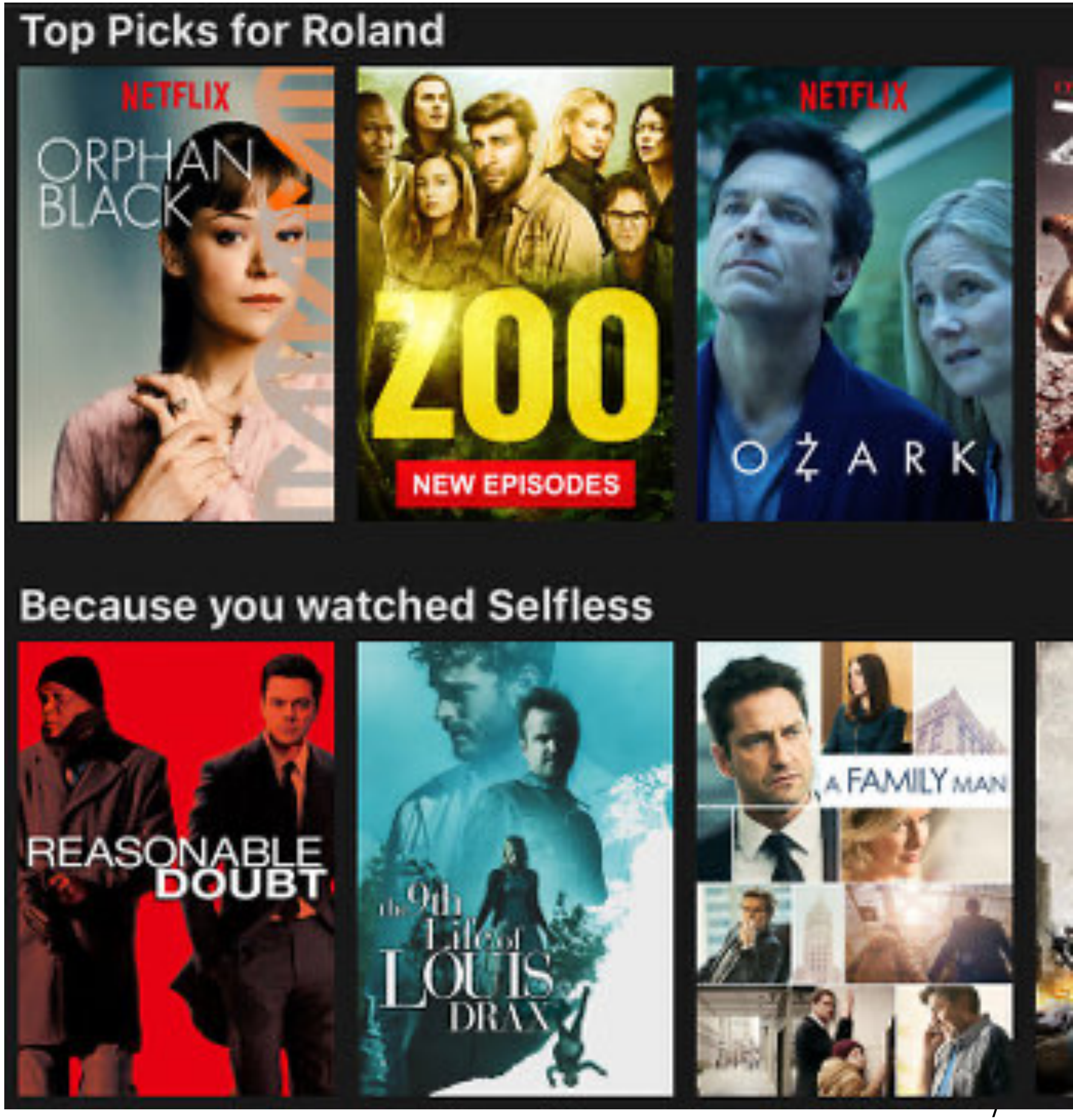
- Increase sales
- Increase customer satisfaction, reduce cancellations
- Opportunity for promotions, persuasion

Recommenders

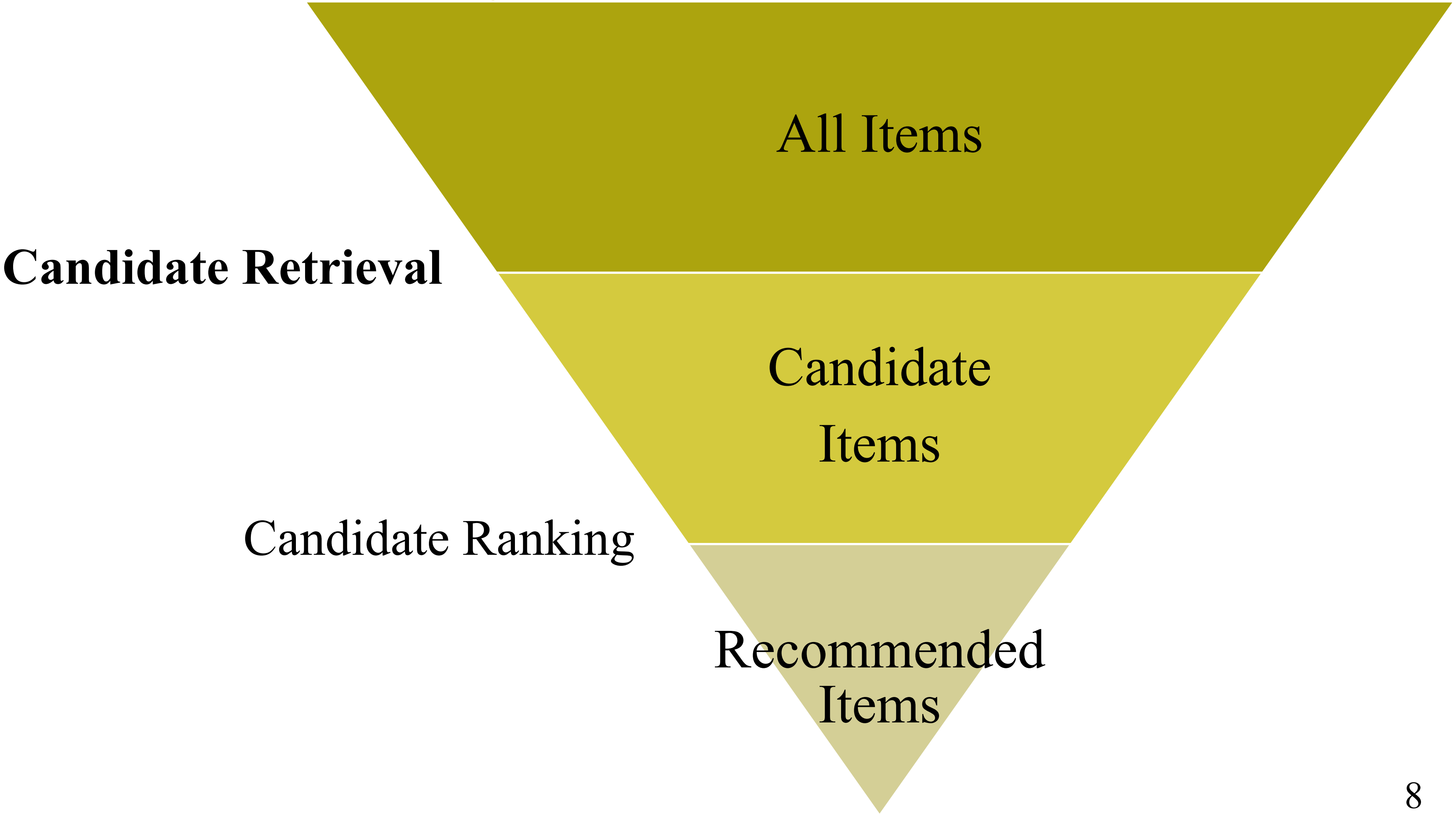
Netflix shows **personalized recommendations** to users.

How would you determine what series and movies to recommend to a user?

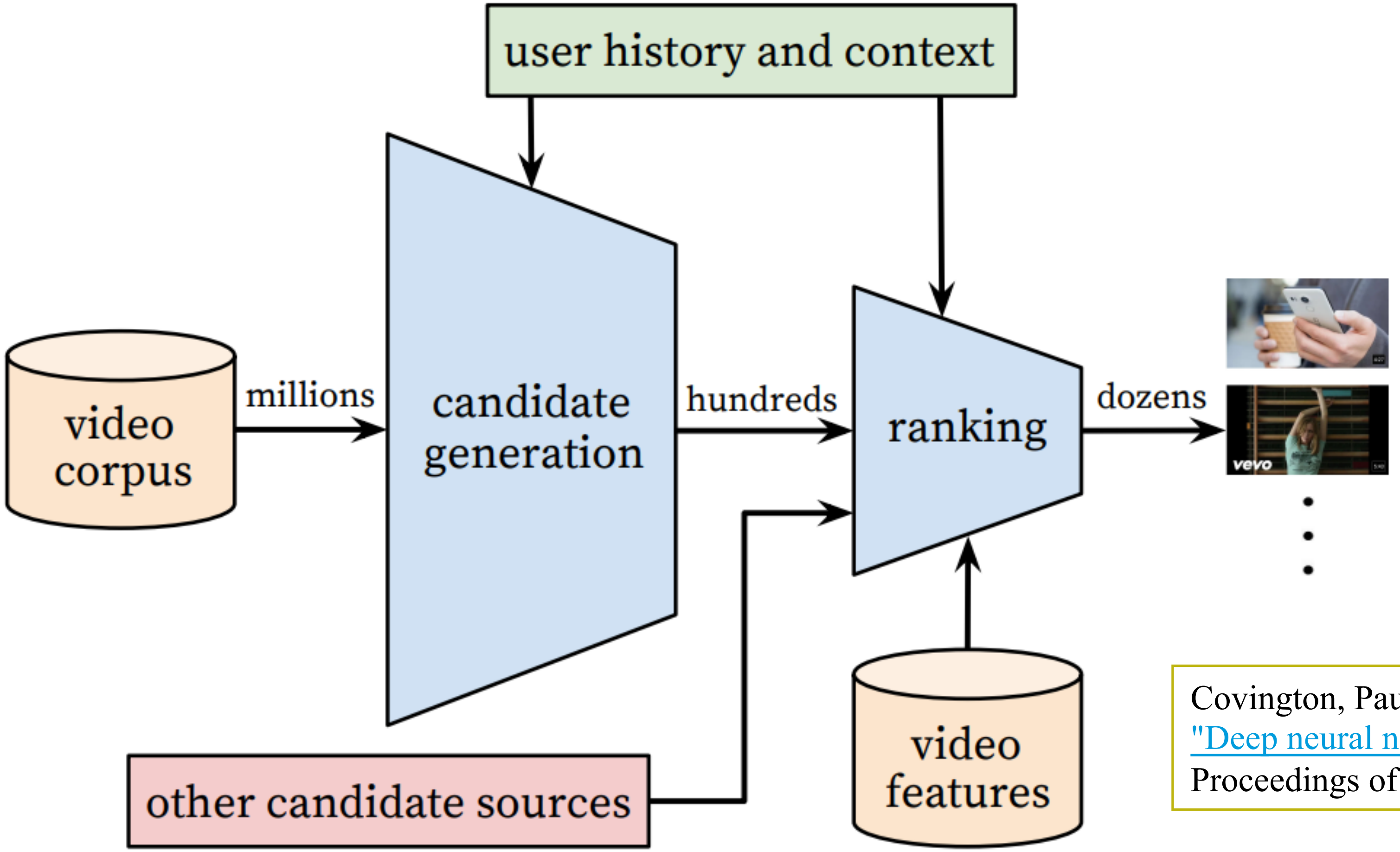
- Which data would you use?
- How would you use that data to determine recommendations?



Large Scale Recommender System Architecture



Recommender System Architecture - YouTube



Covington, Paul, Jay Adams, and Emre Sargin.
["Deep neural networks for YouTube recommendations."]()
Proceedings of the 10th ACM conference on recommender systems, 2016

Candidate Retrieval – Common logic

Find items that are likely to be relevant to the user:

1. Predict ratings for items a user did not see yet
   - Based on other ratings and/or content characteristics of items
2. Select the items with the highest predicted ratings for the user

User Feedback

User feedback is the main input for recommenders.

Explicit feedback

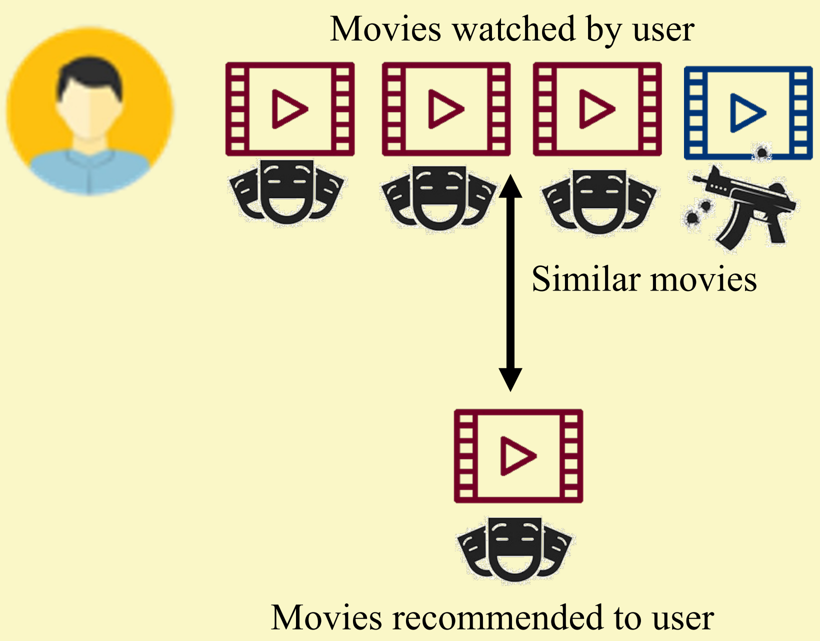
- Numerical rating (e.g., 1-5 stars)
- Binary rating (e.g., like vs. dislike)

Implicit feedback

- Unary rating (e.g., buy, click, view, read, listen, watch)
- Numerical rating (e.g., seconds played)

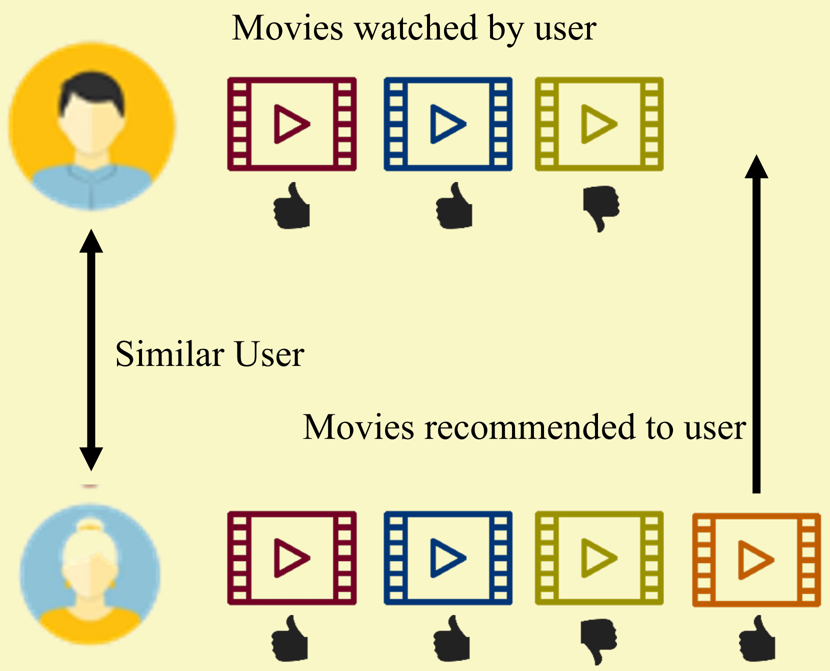
Techniques

Content-based



Similarity based on **items characteristics**: e.g., genre

Collaborative Filtering

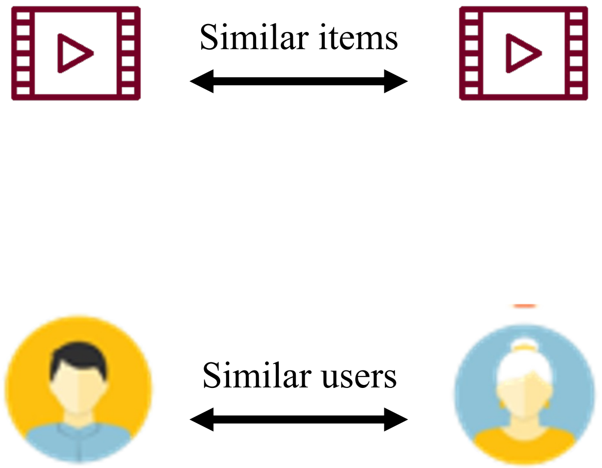


Similarity based on **users taste** (feedback)

Similarity Measures

How to measure similarity between users or items?

- Dice Coefficient
- (Adjusted) Cosine similarity
- Pearson's correlation coefficient



Content-Based Filtering

Content-Based Filtering

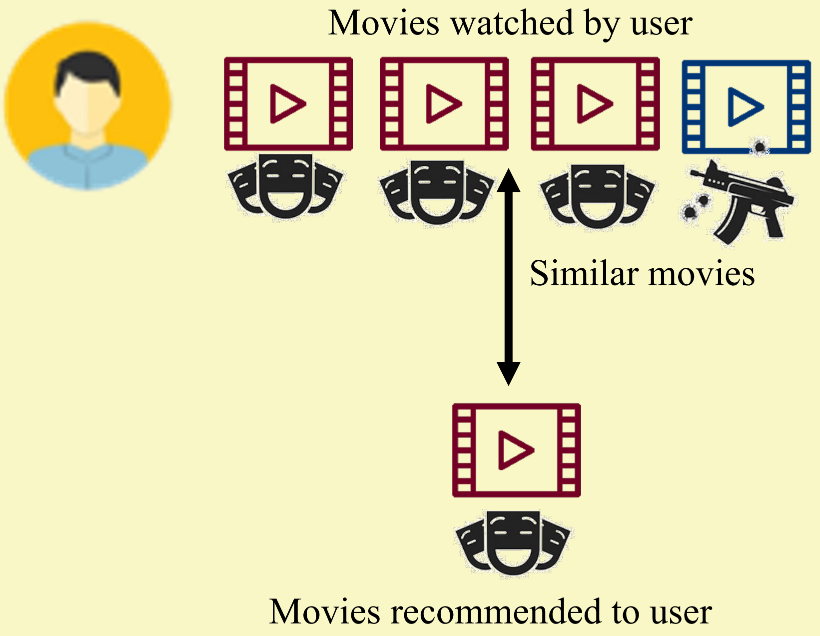
General idea:

- Recommend items that are similar to items the user liked in the past

Required:

- Pre-defined set of item characteristics (e.g., genre, writer, actor)
- Past user behaviour

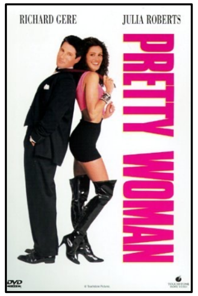
Content-based

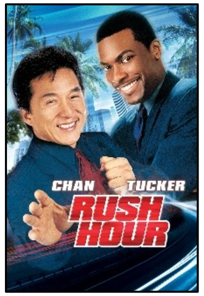


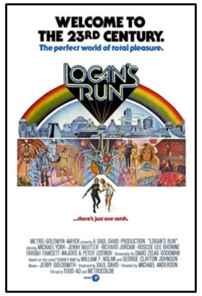
Similarity based on **items characteristics**: e.g., genre

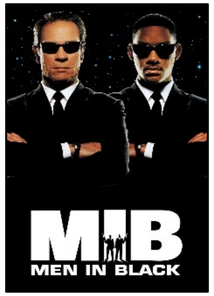
Example 1: Predict if Mike will like MIB

Movie	Action	SF	Comedy	Romance	Children	Mike
Star Wars	x	x				👍
Pretty Woman			x	x		👍
Rush Hour	x		x			👎
101 Dalmatians					x	👎
Logan's Run		x				👎
Men In Black	x	x	x			?









Content-Based Filtering Techniques

- Content-Based Nearest Neighborhood (kNN)
- Probabilistic Approach: Naive Bayes

Content-Based Nearest Neighborhood

1. Determine item similaraties

2. Predict a rating for each unseen item based on rated items that are similar to the unseen item

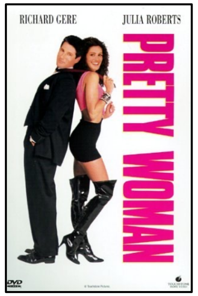
3. Recommend the item with the highest predicted rating

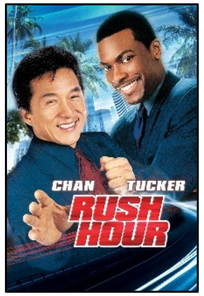
Content-Based Nearest Neighborhood (kNN)

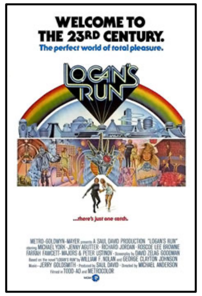
- Based on item characteristics (e.g. genre, writer, actor), calculate the similarities between each pair of items
- For each user u
  - For each unrated item i from user u
    - Construct the set $N_{u,i}$ with the k items that user u has rated that are most similar to i
    - Predict the rating of item i of user u as the average rating over all items in the set $N_{u,i}$
  - Recommend the item i^* to user u that has the highest predicted rating

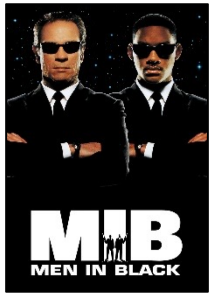
Example 1: Predict if Mike will like MIB

Movie	Action	SF	Comedy	Romance	Children	Mike
Star Wars	x	x				👍
Pretty Woman			x	x		👍
Rush Hour	x		x			👎
101 Dalmatians					x	👎
Logan’s Run		x				👎
Men In Black	x	x	x			?









Similarity Measure

Dice coefficient to compute set similarity:

E.g., Sets *A*,*B* are sets of genres of a movie

$$Dice(A, B) = \frac{2 \cdot |A \cap B|}{|A| + |B|}$$

Example: Star Wars = {Action, SF}

Men In Black = {Action, SF, Comedy}

$$Dice(Star\ Wars, Men\ In\ Black) = \frac{2 \cdot 2}{2 + 3} = \frac{4}{5} = 0.8$$

Example 1: Predict if Mike will like MIB

Dice coefficients:

- *Dice(MIB, Star Wars) = 0.8*
- *Dice(MIB, Pretty Woman) = 0.4*
- *Dice(MIB, Rush Hour) = 0.8*
- *Dice(MIB, 101 Dalmatians) = 0*
- *Dice(MIB, Logan's Run) = 0.5*

Example 1: Predict if Mike will like MIB

Dice coefficients:

- *Dice(MIB, Star Wars) = 0.8*
- *Dice(MIB, Pretty Woman) = 0.4*
- *Dice(MIB, Rush Hour) = 0.8*
- *Dice(MIB, 101 Dalmatians) = 0*
- *Dice(MIB, Logan's Run) = 0.5*

For 2NN:

- N_{MIB} = {Star Wars, Rush Hour}
- P(Like | MIB) = (1+0)/2 = 1/2

For 3NN:

- N_{MIB} = {Star Wars, Rush Hour, Logan's Run}
- P(Like | MIB) = (1+0+0)/3 = 1/3

Example 2: What to recommend to Laura?

- Similarity matrix:

	A	**B**	**C**	**D**	**E**
A	1	0.2	0.4	0.7	0.1
B	0.2	1	0.6	0.3	0.5
C	0.4	0.6	1	0.2	0.7
D	0.7	0.3	0.2	1	0.2
E	0.1	0.5	0.7	0.2	1

- Laura has already consumed items C, D and E and her feedback is:

C	
D	
E	

- Recommendation:

	N_i	**P(Like = 1)**
A		
B		

Example 2: What to recommend to Laura?

- Similarity matrix:

	A	B	C	D	E
A	1	0.2	0.4	0.7	0.1
B	0.2	1	0.6	0.3	0.5
C	0.4	0.6	1	0.2	0.7
D	0.7	0.3	0.2	1	0.2
E	0.1	0.5	0.7	0.2	1

- Laura has already consumed items C, D and E and her feedback is:

C	👍
D	👎
E	👍

2 most similar items (2NN)

- Recommendation:

	N_i	**P(Like = 1)**
A	C,D	
B	C,E	

Example 2: What to recommend to Laura?

- Similarity matrix:

	A	B	C	D	E
A	1	0.2	0.4	0.7	0.1
B	0.2	1	0.6	0.3	0.5
C	0.4	0.6	1	0.2	0.7
D	0.7	0.3	0.2	1	0.2
E	0.1	0.5	0.7	0.2	1

- Laura has already consumed items C, D and E and her feedback is:

C	
D	
E	

- Recommendation:

	N_i	P(Like = 1)
A	C,D	1/2
B	C,E	1

Prediction if Laura will like the item

Example 2: What to recommend to Laura?

- Similarity matrix:

	A	B	C	D	E
A	1	0.2	0.4	0.7	0.1
B	0.2	1	0.6	0.3	0.5
C	0.4	0.6	1	0.2	0.7
D	0.7	0.3	0.2	1	0.2
E	0.1	0.5	0.7	0.2	1

- Laura has already consumed items C, D and E and her feedback is:

C	👍
D	👎
E	👍

Prediction if Laura will like the item

- Recommendation:

	N_i	**P(Like = 1)**
A	C,D	1/2
B	C,E	**1**

Extensions

Similarity-weighted predictions

- P(Like): weigh ratings by similarity (weighted average)
- Example 2: P(Like | A) = (0.4*1 + 0.7*0) / (0.4 + 0.7) = 0.36
- P(Like| B) = (0.6*1 + 0.5*1) / (0.6 + 0.5) = 1

Minimum similarity threshold

- N_i: include only items in Neighbourhood with similarity above some threshold
- Example 2 with threshold = 0.45 : N_A = {D}

  N_B = {C,E}

About Content-Based kNN

Pros:

- Uses item characteristics, so needs limited user data

Cons:

- Not useful for new users (cold start)
- Requires data on item characteristics
- Hard to capture qualitative features such as taste and quality
- Doesn’t use other user’s behaviour
- Depends highly on selected similarity measure
- Calculating similarities does not scale well with number of items

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Probabilistic Approach

Probabilistic Approach (supervised learning)

1. Train a model **per user** on the relation between user feedback and item characteristics
2. Use this model to predict the user rating for unseen items
3. Recommend the item with the highest predicted user rating

- Classification model: Will the user like the item? Probability
- Features: Content based: Genre, writers, actors, etc.
- Algorithms: Naive Bayes, Logistic regression, Decision tree, Random Forest, Neural Network, etc.

Probabilistic Approach - Challenges

Train a model **per user** on the relation between user feedback and item characteristics

Data:

- Typical users have only rated a few items: data is scarce

Performance:

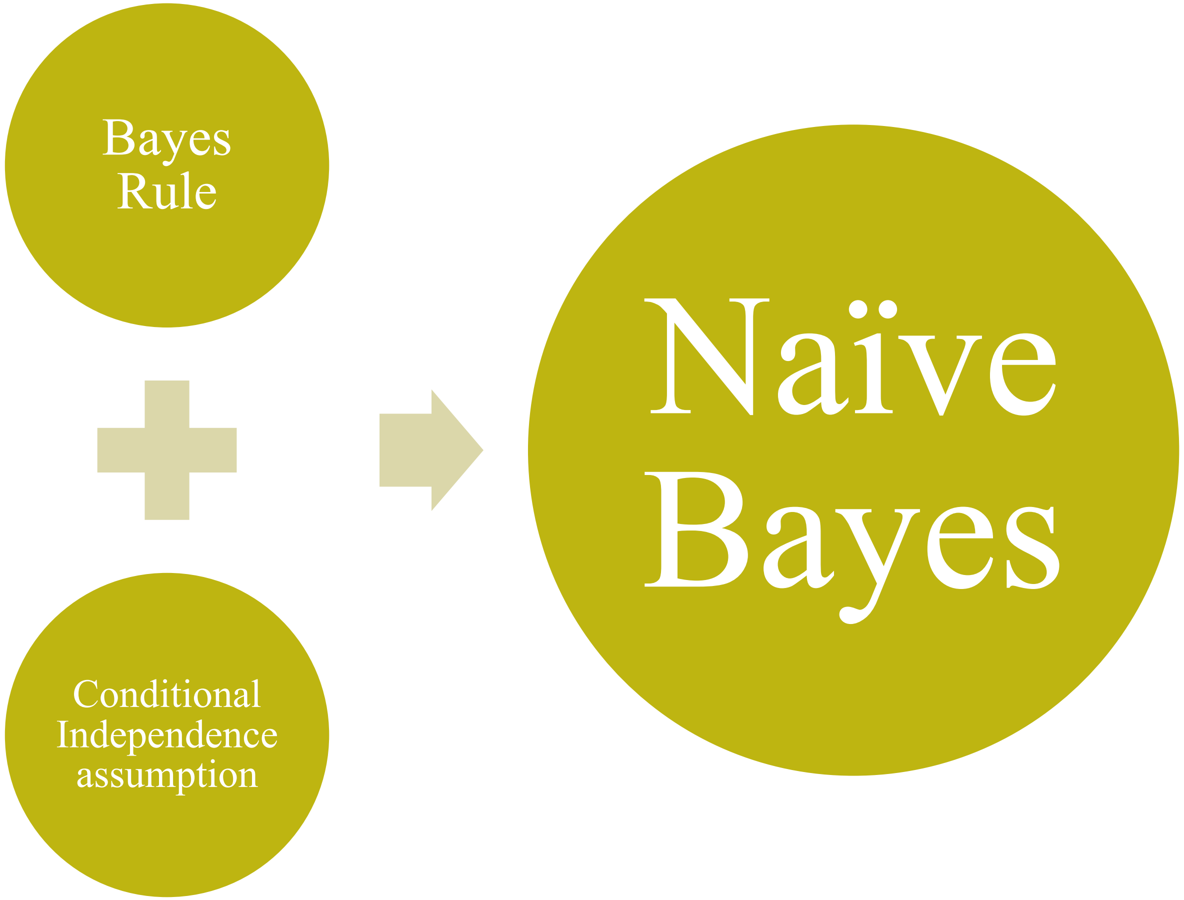
- We have to estimate many models, so estimation has to be fast

➢ **Naïve Bayes** is a popular algorithm because it performs relatively well with few observations and is fast to estimate

Naïve Bayes for Content Based Reccomendations

- Outcome: Y = {Like, Not Like}
- Features: $X = (X_1, X_2, X_3, \ldots, X_k)$, e.g.,
  - Genre = {action, comedy, …}
  - Actor = {Bruce Willis, ..}
  - ..
- We are interested in estimating

$$\mathrm{P}(Y = \text{Like} \mid X_1 = x_1, X_2 = x_2, \ldots, X_k = x_k)$$



Example 1: Predict if Mike will like MIB

Movie	Action	SF	Comedy	Romance	Children	Mike
Star Wars	x	x				👍
Pretty Woman			x	x		👍
Rush Hour	x		x			👎
101 Dalmatians					x	👎
Logan’s Run		x				👎
Men In Black	x	x	x			?

Predict if Mike will like a movie that is a combination of an Action, SF and comedy:
P (Like | Action & SF & Comedy & NOT Romance & NOT Children)

Bayes' Rule

Definition Conditional Probability:
$P(A|B) = P(A \cap B) / P(B)$

$$P(A|B) = P(B|A) \times \frac{P(A)}{P(B)}$$

Romance	1	0
NOT Romance	1	3

$P(\text{Like} \mid \text{Romance}) = P(\text{Romance} \mid \text{Like}) \cdot P(\text{Like})/P(\text{Romance})$

$= 0.5 \cdot 0.4/0.2 = 1$

Example 1: Bayes' Rule

Movie	Action	SF	Comedy	Romance	Children	Mike
Star Wars	x	x				
Pretty Woman			x	x		
Rush Hour	x		x			
101 Dalmatians					x	
Logan's Run		x				
Men In Black	x	x	x			?

P(Like | Action & SF & Comedy & NOT Romance & NOT Children) =
P(Action & SF & Comedy & NOT Romance & NOT Children | Like) *
P(Like) / P(Action & SF & Comedy & NOT Romance & NOT Children)

Example 1: Bayes' Rule

Movie	Action	SF	Comedy	Romance	Children	Mike
Star Wars	x	x				👍
Pretty Woman			x	x		👍
Rush Hour	x		x			👎
101 Dalmatians					x	👎
Logan's Run		x				👎
Men In Black	x	x	x			?

P(Like | Action & SF & Comedy & NOT Romance & NOT Children) =
P(Action & SF & Comedy & NOT Romance & NOT Children | Like) *
P(Like) / P(Action & SF & Comedy & NOT Romance & NOT Children) = **0** * 0.4 / **0**

No rated movie is a combination of Action, SF and comedy

Example 1: Naïve Bayes

Movie	Action	SF	Comedy	Romance	Children	Mike
Star Wars	x	x				👍
Pretty Woman			x	x		👍
Rush Hour	x		x			👎
101 Dalmatians					x	👎
Logan's Run		x				👎
Men In Black	x	x	x			?

Mikes ratings for Action movies
Mikes ratings for SF movies
Mikes ratings for Comedies
Mikes ratings for <u>NOT</u> Romantic movies
Mikes ratings for <u>NOT</u> Children movies

Combine observations
(**assuming independence**)

Naive Bayes

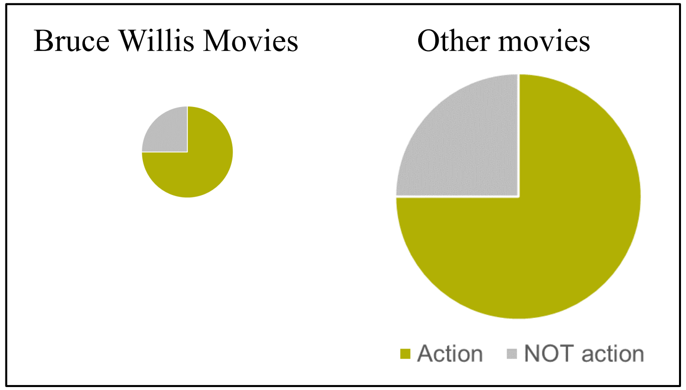
Assumption: Given the outcome of the trial, the value of attribute 1 is independent of the value of attribute 2

Example:

- Attribute 1: Action
- Attribute 2: Bruce Willis
- Assumption: P(Action | Like) = P(Action | Like & Bruce Willis)

  = P(Action | Like & NOT Bruce Willis)

Liked movies:



And thus: **P(Action & Bruce Willis | Like) = P(Action | Like) * P(Bruce Willis | Like)**

Note: We don’t assume that the probability of a Like is the same for Action and Bruce Willis, but that the fraction of action movies within all liked movies is the same among Bruce Willis movies and the rest of the movies.

Naïve Bayes for Reccomendations

$$\mathbf{P}(\boldsymbol{Y} = \mathbf{Like} \mid \boldsymbol{X_1} = \boldsymbol{x_1}, \boldsymbol{X_2} = \boldsymbol{x_2}, \dots, \boldsymbol{X_k} = \boldsymbol{x_k}) =$$

$$\frac{\mathrm{P}(Y = \mathrm{Like})\, \mathrm{P}(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k \mid Y = \mathrm{Like})}{\mathrm{P}(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k)} =$$

$$\frac{\mathrm{P}(Y = \mathrm{Like})\, \mathrm{P}(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k \mid Y = \mathrm{Like})}{\sum_y \mathrm{P}(Y = \mathrm{y})\, \mathrm{P}(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k \mid Y = \mathrm{y})}$$

$$\mathrm{P}(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k \mid Y = \mathrm{y}) =$$
$$\mathrm{P}(X_1 = x_1 \mid Y = \mathrm{Like})\, \mathrm{P}(X_2 = x_2 \mid Y = \mathrm{y}) \ \dots =$$
$$\prod_{i=1}^{k} \mathrm{P}(X_i = x_i \mid Y = \mathrm{y})$$

Bayes Rule

Conditional Independence assumption

Naïve Bayes

$$\mathbf{P}(\boldsymbol{Y} = \mathbf{Like} \mid \boldsymbol{X_1} = \boldsymbol{x_1}, \boldsymbol{X_2} = \boldsymbol{x_2}, \dots, \boldsymbol{X_k} = \boldsymbol{x_k}) =$$

$$\frac{\mathbf{P}(\boldsymbol{Y} = \mathbf{Like}) \prod_{i=1}^{k} \mathbf{P}(\boldsymbol{X_i} = \boldsymbol{x_i} \mid \boldsymbol{Y} = \mathbf{Like})}{\sum_y \mathbf{P}(\boldsymbol{Y} = \mathbf{y}) \prod_{i=1}^{k} \mathbf{P}(\boldsymbol{X_i} = \boldsymbol{x_i} \mid \boldsymbol{Y} = \mathbf{y})}$$

Example 1: Naïve Bayes

P(Like | Action & SF & Comedy & NOT Romance & NOT Children) =

P(Like) *
P(Action | Like) * P(SF | Like) * P(Comedy | Like) *
P(NOT Romance | Like) * P(NOT Children | Like)

[P(Like) *
P(Action | Like) * P(SF | Like) * P(Comedy | Like) *
P(NOT Romance | Like) * P(NOT Children | Like)
+
P(NOT Like) *
P(Action | NOT Like) * P(SF | NOT Like) * P(Comedy | NOT Like) *
P(NOT Romance | NOT Like) * P(NOT Children | NOT Like)]

Example 1:

Likes

P(Like) = 2/5

P(Action | Like) = 1/2

P(SF | Like) = 1/2

P(Comedy | Like) = 1/2

P(NOT Romance | Like) = 1/2

P(NOT Children | Like) = 2/2

All multiplied = 0.025

Dislikes

- P(NOT Like) = 3/5
- P(Action | NOT Like = 0) = 1/3
- P(SF | NOT Like) = 1/3
- P(Comedy | NOT Like) = 1/3
- P(NOT Romance | NOT Like) = 3/3
- P(NOT Children | NOT Like) = 2/3

All multiplied = 0.015

- P(Like | MIB) = 0.025 / (0.025 + 0.015) = 0.628

About Naive Bayes

Pros:

- Better accuracy than most other probabilistic models when observations are limited
- Faster to estimate than most other probabilistic models

Cons:

- Not useful for new users (cold start)
- Requires data on item characteristics
- Hard to capture qualitative features such as taste and quality
- Doesn’t use other user’s behaviour, requires to train a seperate model for each user

Naïve Bayes vs. Content-Based NN

Content-Based Nearest Neighorhood (kNN)

- Use the k items most similar to item i to predict if the user will like i
- Similarity is based on item characteristics

Naive Bayes

- Use the characteristics of item i to predict if the user will like i
- Each characteristic is considered independently

Naïve Bayes vs. Content-Based NN

Content-Based Nearest Neighorhood (kNN)

- If I like "Mission Impossible" then my predicted rating for "Minority Report" might go up, but not for "Eyes Wide Shut". "Eyes Wide Shut" features Tom Cruise, but is not very similar to "Mission Impossible" considering other characteristics.
- Captures "local" similarities

Naïve Bayes

- If I like "Mission Impossible" then my predicted ratings for all action movies and for all Tom Cruise movies go up
- Captures "global" patterns

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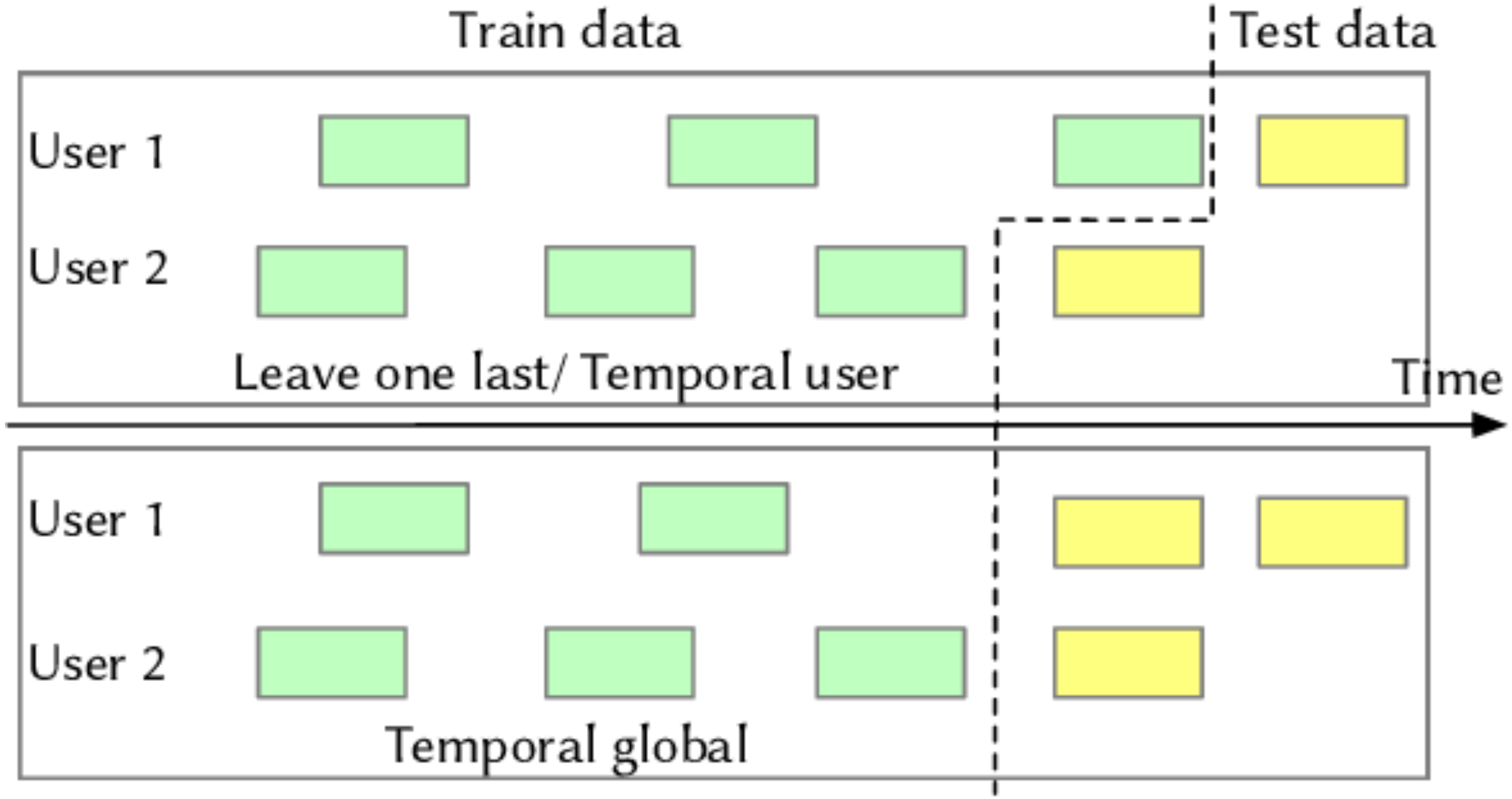
Evaluation of Recommenders

Evaluation

	Offline evaluation	Online evaluation
How?	Split data	Experimentation: A/B test. Online models (reinforcement learning)
Performance Metrics	Model prediction metrics, e.g., MSE	Business metrics: e.g., clicks, sales, customer loyalty
Qualititative metrics	Serendipity, variety	
Main Challenge	Recommendations are not actually shown to user -> Bias	Position bias Short term vs long term performance

Offline Evaluation: split data

- Random vs temporal
- Users
  - Exclude Cold start users



From: Exploring Data Splitting Strategies for the Evaluation of Recommendation Models; Meng, McCreadie, Macdonald, Ounis

Model Evaluation metrics

- Error of predicted ratings
  - MAE, RMSE (continuous rating)
  - Precision, Recall (binary ratings)

But… only a few predictions (Top k) will be actually used for recommendations

- Evaluate how good the k recommended items are:
  - Recall @ k = $\frac{tp}{tp+fn} = \frac{|good\ recommendations|}{|\ all\ good\ items|}$
  - Precision @ k = $\frac{tp}{tp+fp} = \frac{|good\ recommendations|}{|\ all\ recommendations\ |}$

Model Evaluation metrics - example

Item	Actual Rating	Model 1 prediction	Model 2 Prediction
A	5	4.0	4.0
B	4	3.4	4.9
C	4	3.3	4.8
D	4	2.9	2.0
E	4	2.3	1.9
F	3	4.1	2.2
G	3	4.5	2.5
H	3	2.3	1.2
I	2	2.2	3.4
J	2	1.9	3.4

MSE:

- **Model 1: 1.0**
- Model 2: 1.9

Model Evaluation metrics - example

Item	Actual Rating	Model 1 prediction	Model 2 Prediction
A	5	**4.0**	**4.0**
B	4	3.4	**4.9**
C	4	3.3	**4.8**
D	4	2.9	2.0
E	4	2.3	1.9
F	3	**4.1**	2.2
G	3	**4.5**	2.5
H	3	2.3	1.2
I	2	2.2	3.4
J	2	1.9	3.4

MSE:

- **Model 1: 1.0**
- Model 2: 1.9

Recall @ 3 $= \frac{|good\ recommendations|}{|\ all\ good\ items|}$

- Model 1: 1/5
- **Model 2: 3/5**

Precision @ 3 $= \frac{|good\ recommendations|}{|\ all\ recommendations\ |}$

- Model 1: 1/3
- **Model 2: 3/3**

Qualitative metrics - Serendipity

- "More of the same"
- A trap for any recommender, especially content-based

Example:

Qualitative metrics: What recommendations are preferred?

Recommendations from the long tail:

Recommend unkown items that users will actually like

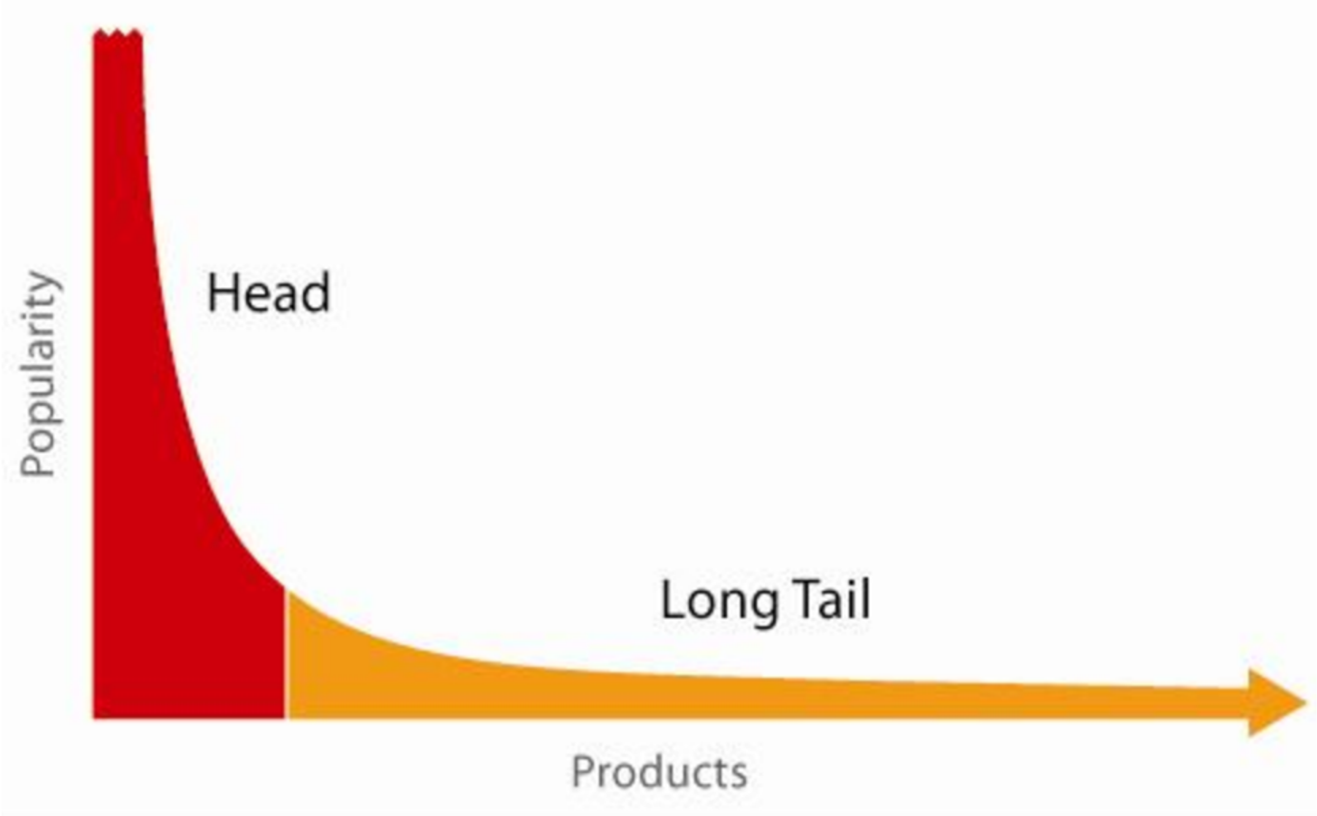
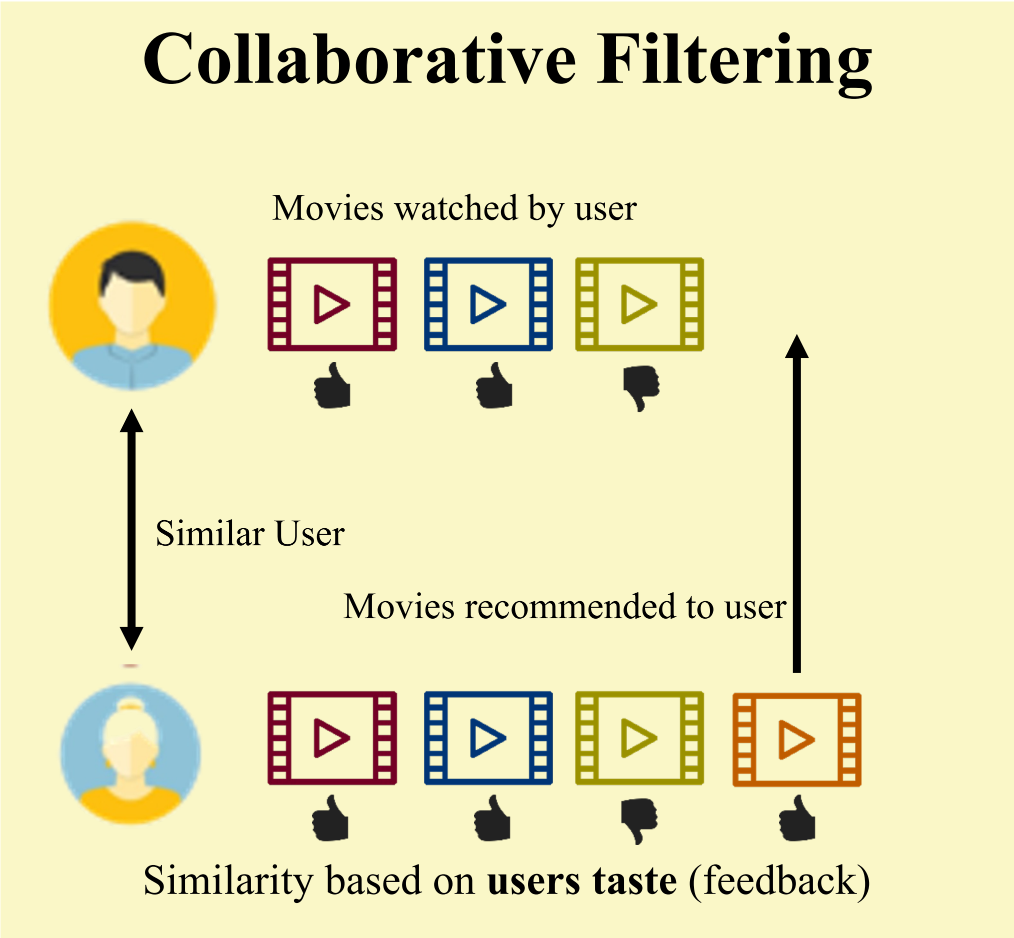


Image taken from: https://susanajorgina.wordpress.com/tag/the-long-tail/

Next week: Collaborative Filtering



Today’s Course Material

- Lecture notes
- Exercises